

Yingbo Fan

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Research Profile

I develop multimodal spatial intelligence that enables embodied systems to perceive complex environments, predict their dynamics, and act reliably. My research connects panoramic and 3D perception, efficient world models, and real-world deployment in intelligent mining.

Research interests: Multimodal Spatial Intelligence - World Models - Intelligent Mining

Academic Record

8 Peer-reviewed journal articles 9 publications in total	5 First-author publications Computer vision and AI	16 Granted patents China and overseas jurisdictions	5 Major research projects National and ministerial programs
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Current Research

Research theme	Efficient Multiview Embodied Reasoning Geometry-aware perception and action under constrained computation Multiview spatial grounding - Efficient multimodal reasoning - Perception-to-action alignment
Research theme	Efficient Predictive World Models Planning-capable representations with lower inference cost Compact predictive representations - Planning-oriented efficiency - Deployment-aware evaluation
Research theme	Foundation-Model-Based Panoramic Geometry Geometrically coherent spatial perception for 360-degree vision Panoramic scene geometry - Foundation-model adaptation - Spatially coherent perception

Experience

2026-Present	Postdoctoral Researcher Peking University
2016	Visiting Researcher The Hong Kong Polytechnic University

Education

2021-2025	Ph.D. in Cartography and Geographic Information Systems Peking University
2018-2021	M.S. in Computer Science and Technology Southeast University
2014-2018	B.E. in Electrical Engineering and Automation University of Electronic Science and Technology of China

Publications

Adaptive Multi-Channel Dehazing for Enhanced Visibility in Underground Coal Mine Images

Yingbo Fan, Shanjun Mao, Mei Li, Boxiang Yang, Yinglu Yang

PLOS ONE, 20(11): e0334251 - DOI: [10.1371/journal.pone.0334251](https://doi.org/10.1371/journal.pone.0334251)

Reflections on the Development of Advanced Intelligent Coal Mines

Shanjun Mao, Huazhou Chen, Haibin Liu, et al., Yingbo Fan, et al.

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CM-YOLOv8: Lightweight YOLO for Coal Mine Fully Mechanized Mining Face

Yingbo Fan, Shanjun Mao, Mei Li, Zheng Wu, Jitong Kang

Sensors, 24(6): 1866 - DOI: [10.3390/s24061866](https://doi.org/10.3390/s24061866)

Rectification for Stitched Images with Deformable Meshes and Residual Networks

Yingbo Fan, Shanjun Mao, Mei Li, Zheng Wu, Jitong Kang, Ben Li

Applied Sciences, 14(7): 2821 - DOI: [10.3390/app14072821](https://doi.org/10.3390/app14072821)

Experimental Study on Drilling Signal Characteristics of PDC Drill Bit in Media of Different Strengths and Identification of Weak Media

Zheng Wu, Yingbo Fan, Huazhou Chen

Sensors, 24(23): 7852 - DOI: [10.3390/s24237852](https://doi.org/10.3390/s24237852)

A Coal Mine Tunnel Deformation Detection Method Using Point Cloud Data

Jitong Kang, Mei Li, Shanjun Mao, Yingbo Fan, Zheng Wu, Ben Li

Sensors, 24(7): 2299 - DOI: [10.3390/s24072299](https://doi.org/10.3390/s24072299)

A Point Cloud Segmentation Method for Dim and Cluttered Underground Tunnel Scenes Based on the Segment Anything Model

Jitong Kang, Ning Chen, Mei Li, Shanjun Mao, Haoyuan Zhang, Yingbo Fan, Hui Liu

Remote Sensing, 16(1): 97 - DOI: [10.3390/rs16010097](https://doi.org/10.3390/rs16010097)

LMFD: Lightweight Multi-Feature Descriptors for Image Stitching

Yingbo Fan, Shanjun Mao, Mei Li, Jitong Kang, Ben Li

Scientific Reports, 13: 21162 - DOI: [10.1038/s41598-023-48432-7](https://doi.org/10.1038/s41598-023-48432-7)

HF PQ: Deep Neural Network Compression by Hardware-Friendly Pruning-Quantization

Yingbo Fan, Wei Pang, Shengli Lu

Applied Intelligence, 51(10): 7016-7028 - DOI: [10.1007/s10489-020-01968-x](https://doi.org/10.1007/s10489-020-01968-x)

Selected Major Research Programs

2025-2030	Transparent Mine Geological Foundation Model and Cloud Platform National Science and Technology Major Project - Research contributor Contributing to intelligent processing for heterogeneous video streams, 3D spatial data, scene reconstruction, and multimodal fusion.
2024-2026	Vector-Space Video Streaming for Intelligent Longwall Mining Faces Science and Technology Support Project - Key research contributor Research on panoramic video streams, spatial mapping, and real-time visual interaction for an intelligent longwall mining platform.
2023-2026	Digital-Twin Mining Area and Decision System for Deep-Sea Polymetallic Nodules National Key Research and Development Program - Research contributor Contributing to underwater scene reconstruction and digital-twin technologies for deep-sea mining environments.
2022-2025	Digital-Twin Technologies for Coal-Mine Disaster Monitoring and Decision-Making National Key Research and Development Program - Research contributor Research on depth enhancement and local 3D reconstruction for improving digital-twin scene fidelity.

2020-2023 High-Precision Perception and Digital-Twin Systems for Large Mine Roadway Robots
 National Key Research and Development Program - Research contributor
 Research on multi-source sensing, spatial perception, and 3D digital-twin construction for intelligent roadway systems.

Awards

2024	Liao Kaiyuan Scholarship, Peking University
2024	Academic Excellence Award, Peking University
2023	Social Practice Excellence Award, Peking University
Selected	Outstanding Young Paper Award, National Virtual Geographic Environment Conference
2021	Outstanding Presentation Award, International Graduate Workshop on GeoInformatics

Granted Patents

2025	Intelligent Multidimensional-Code Parsing Method and System for Visual-Information Processing Results China - ZL 2024 1 1409062.8
2025	Equipment Control System from Geographic Space to High-Precision Digital Space China - ZL 2023 1 1396230.X
2024	Method for Constructing a General Artificial Spatiotemporal Intelligence Foundation Model China - ZL 2024 1 1433513.1
2024	Image Rectangling Method and Apparatus Based on Deformable Meshes and Wide Residual Networks China - ZL 2024 1 0222860.3
2024	Method and Apparatus for Vector-Space Construction and Image-Fusion Modeling in Confined Spaces China - ZL 2023 1 1577506.4
2024	Artificial Intelligence System Based on Spatiotemporal Information Pairs China - ZL 2024 1 0643385.7
2024	Multi-View Geometric Scene Construction Method and Apparatus for Confined Spaces China - ZL 2023 1 1058030.3
2023	Method and Apparatus for Real-Time Construction of Real Geographic-Space Scenes China - ZL 2022 1 0499659.0
2024	Image Feature Matching Method and Apparatus Based on Multiple Feature Descriptors China - ZL 2023 1 0841374.5
2023	Method and Apparatus for Constructing Real Geographic Space Scene in Real Time United States - US 11,836,878 B2
2024	Method and Apparatus for Constructing Real Geographic Space Scene in Real Time Russia - RU 2815084
2024	Method and Apparatus for Constructing a Scene in Real Geographic Space in Real Time Canada - CA 3199062
2023	Method and Apparatus for Constructing Real-Geographic-Space Scene in Real Time Australia - AU 2023202859
2025	Artificial Intelligence System Based on Spatio-Temporal Information Pairs Russia - RU 2842579
2023	Face Detection Method Based on the AdaBoost Algorithm China - ZL 2019 1 1052238.6
2021	Deep Neural Network Hardware Accelerator Based on Power-of-Two Quantization China - ZL 2019 1 1436775.1